

Multi-Modal Agentic AI System for Medical Diagnosis and Triage

Project Timeline

1. Project Overview

This document presents the full project timeline for the Multi-Modal Agentic AI System for Medical Diagnosis and Triage. The system autonomously diagnoses chest and cardiovascular conditions by combining three machine learning models, a medical large language model (MedGemma-27B-IT), and a retrieval-augmented generation (RAG) layer. The pipeline processes chest X-ray images, structured electronic health records (EHR), and free-text clinical notes, and delivers a triage decision with specialist referral recommendation.

2. Architecture Summary

The system comprises three ML models, one agent LLM, and one RAG layer:

Component	Technology / Model	Type	Purpose
Model 1 — ChestVision	ResNet-50 / ViT-B/16 (from scratch)	Custom Neural Network	Multi-label X-ray classification (14 conditions) → ONNX
Model 2 — ClinicalFusion	XGBoost + OpenMed NER encoder fusion	Hybrid Model	Diagnosis (5 categories) + Severity (0–3) → ONNX
Model 3 — OpenMed NER	OpenMed-NER-DiseaseDetect-SuperClinical-434M	Pretrained + Fine-tuned	Biomedical NER on clinical notes → ONNX
Agent LLM	google/medgemma-27b-it	Instruction-tuned LLM	Final diagnosis reasoning via HuggingFace Inference API
RAG Layer	A-Z Medical Encyclopedia + ChromaDB	Post-decision QA	Supporting context enrichment after diagnosis
Application	Next.js 14 + LangGraph.js + Clerk + NeonDB	Full-stack app	Patient-facing UI, agent orchestration, appointments

3. Phase-by-Phase Timeline

Phase 1: Data Preparation | 20 May – 2 Jun 2026 | Weeks 1 – 2

Objective: Establish clean, split data pipelines for all three models. No model training starts until data is verified and processed.

Task	Dates	Status	Deliverable
NIH ChestX-ray14 Preprocessing	20–26 May	Complete	train/val/test CSVs (85,108 images)
Synthea EHR Feature Engineering	20–26 May	Complete	Scaled feature matrix + label encoder
BIO-tag NER Dataset (Synthea Notes)	27 May–2 Jun	Pending	BIO-tagged NER dataset for Model 3 fine-tuning
Medical Encyclopedia → ChromaDB Setup	3–9 Jun	Pending	Populated ChromaDB collection (RAG)

Phase 2: Model 3 — OpenMed NER (Pretrained + Transfer Learning) | 27 May – 16 Jun 2026 | Weeks 2 – 4

Objective: Fine-tune the OpenMed-NER-DiseaseDetect-SuperClinical-434M pretrained model on Synthea chest/cardiovascular notes. This model must be completed first as its encoder feeds directly into Model 2 training.

Task	Dates	Status	Deliverable
OpenMed NER Fine-tuning	27 May–9 Jun	Pending	Fine-tuned NER weights on Synthea notes
NER Evaluation (seqeval F1)	10–12 Jun	Pending	Entity F1 ≥ 0.82 ; transfer gain ≥ 3 F1 pts
ONNX Export — NER Model (INT8)	13–16 Jun	Pending	Quantised ONNX model ~150MB

Phase 3: Model 2 — ClinicalFusion (Hybrid Model) | 10 Jun – 1 Jul 2026 | Weeks 3 – 6

Objective: Build the hybrid XGBoost + OpenMed NER encoder fusion model trained on Synthea EHR data. Produces diagnosis classification (5 categories) and severity scoring (0–3).

Task	Dates	Status	Deliverable
XGBoost Tabular Baseline Training	10–17 Jun	Pending	XGBoost baseline; GridSearchCV results
Fusion Network + OpenMed Integration	17 Jun–1 Jul	Pending	Trained fusion model; ablation table
ClinicalFusion Evaluation & Ablation	24–28 Jun	Pending	Macro F1 ≥ 0.80 ; false-low rate $< 2\%$
ONNX Export — ClinicalFusion	29 Jun–1 Jul	Pending	Validated ONNX pipeline ~120MB

Phase 4: Model 1 — ChestVision (Custom Neural Network) | 17 Jun – 15 Jul 2026 | Weeks 4 – 8

Objective: Train a custom ResNet-50 (and optionally ViT-B/16) from scratch on NIH ChestX-ray14 for multi-label thoracic condition classification. GPU-intensive — scheduled to run overnight in parallel with application build.

Task	Dates	Status	Deliverable
ResNet-50 / ViT-B/16 Training	17 Jun–8 Jul	Pending	Best checkpoint; training curves; Optuna report
Grad-CAM & Explainability	8–10 Jul	Pending	Heatmap overlays; localisation AUC
ChestVision Evaluation (AUC-ROC)	8–11 Jul	Pending	Mean AUC ≥ 0.82 ; per-class AUC ≥ 0.75
ONNX Export + INT8 Quantisation (All Models)	11–15 Jul	Pending	All 3 models as validated ONNX files $< 250\text{MB}$

Phase 5: Application Build | 17 Jun – 19 Jul 2026 | Weeks 5 – 9

Objective: Build the full-stack Next.js 14 application with Clerk authentication, NeonDB, LangGraph.js agent, and MedGemma-27B-IT integration. Runs in parallel with model training phases.

Task	Dates	Status	Deliverable
Next.js 14 + Clerk + NeonDB Scaffold	17 Jun–1 Jul	Pending	Auth-gated app with DB schema and routes
ONNX Inference API Routes (×3)	1–8 Jul	Pending	3 working model inference endpoints

Task	Dates	Status	Deliverable
LangGraph Agent + MedGemma-27B-IT	8–15 Jul	Pending	11-node stateful diagnostic agent
ChromaDB RAG Integration	8–12 Jul	Pending	Top-3 encyclopedia retrieval wired to agent
Google Calendar + Gmail API	12–19 Jul	Pending	Appointment booking and email confirmation flow

Phase 6: Production Deployment | 15 – 22 Jul 2026 | Weeks 9 – 10

Objective: End-to-end testing and production deployment to Vercel. CI/CD pipeline via GitHub Actions. All ONNX models validated within Vercel 250MB function size limits.

Task	Dates	Status	Deliverable
E2E Testing & QA (Playwright / Jest)	15–19 Jul	Pending	Smoke test with synthetic patient scenario
Vercel Deployment + CI/CD Pipeline	19–22 Jul	Pending	Live production URL; automated ONNX validation

Phase 7: Report Writing & Submission | 15 – 24 Jul 2026 | Weeks 9 – 10

Objective: Final report writing, presentation preparation, and project submission. Runs in parallel with production deployment.

Task	Dates	Status	Deliverable
Report Writing & Presentation Prep	15–24 Jul	Pending	Final report + presentation slides submitted by 24 Jul

4. Key Milestones

Milestone	Target Date	Project Week
Phase 1 Complete	2 Jun 2026	End of Week 2
OpenMed NER ONNX Export Ready	16 Jun 2026	End of Week 4
ClinicalFusion Model Validated	1 Jul 2026	End of Week 6
All 3 ONNX Models Quantised & Ready	15 Jul 2026	End of Week 8
Full Agent End-to-End Test Passing	19 Jul 2026	Week 9
Production Deployment Live (Vercel)	22 Jul 2026	Week 10
Final Report & Project Submission	24 Jul 2026	Week 10

5. Project Gantt Chart

The Gantt chart below illustrates the full 10-week schedule, showing task dependencies, parallel workstreams, and the production deployment phase immediately before the submission deadline.

